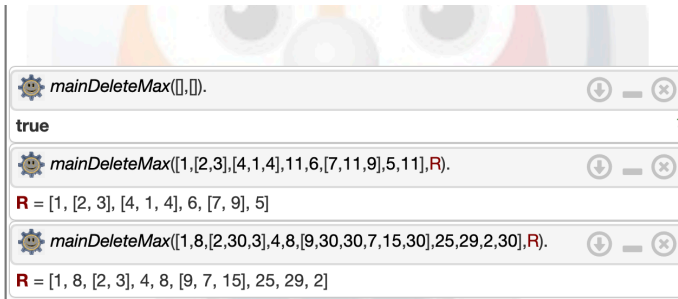


Test Prolog

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```
%PB7
%testele:
%a) []
%b) [1,[2,3],[4,1,4],11,6,[7,11,9],5,11]
%c) [1,8,[2,30,3],4,8,[9,30,30,7,15,30],25,29,2,30]

% maxElem(L:list, R:number)
% (i, o)
%model matematic:
% maxElem(l1...ln) =
%= l1, daca n = 1 si l1 este numar
%= maxElem(l1), daca n = 1 și l1 este lista
%=max(l1, maxElem(l2...ln)), daca l1 este nr
%=max(maxElem(l1), maxElem(l2...ln)), daca l1 este lista

mymax(A,B,B) :-
    A <= B, !.
mymax(A,_,A).

maxElem([H],H) :-
    number(H), !.
maxElem([H],R) :-
    is_list(H), !,
    maxElem(H,R).

maxElem([H|T],R) :-
    number(H),
    !,
    maxElem(T,RT),
    mymax(H,RT,R).

maxElem([H|T],R) :-
    is_list(H),
    !,
```

```

maxElem([H|T],R) :-
    is_list(H),
    !,
    maxElem(H,RH),
    maxElem(T,RT),
    mymax(RH,RT,R).

|

%model mathematic:
% deleteMax(l1...ln, M) =
%   = [], daca n = 0
%   =deleteMax(l2...ln, M), daca l1 = M
% =deleteMax(l1, M) U deleteMax(l2...ln, M), daca l1 este lista
% = {l1} u deleteMax(l2...ln, M), altfel
%
% deleteMax(L:list, M:number, R:list)
% (i, i, o)

deleteMax([],_,[]).

deleteMax([H|T],M,R) :-
    number(H),
    H == M,
    !,
    deleteMax(T,M,R).

deleteMax([H|T],M,[RH|R]) :-
    is_list(H),
    !,
    deleteMax(H,M,RH),
    deleteMax(T,M,R).

deleteMax([H|T],M,[H|R]) :-
    deleteMax(T,M,R).

% mainDeleteMax(L:list, R:list)
% (i, o)
mainDeleteMax([],[]).
mainDeleteMax(L,R) :-
    maxElem(L,M),
    deleteMax(L,M,R).

% mainDeleteMax([1,[2,3],[4,1,4],11,6,[7,11,9],5,11],R).
% mainDeleteMax([1,8,[2,30,3],4,8,[9,30,30,7,15,30],25,29,2,30],R).

```